4. When these replacements are made, the Hamiltonian becomes an operator that can be used in either the time-dependent or time-independent Schrödinger equation. The time-dependent equation tells us how the wave function changes with time. The time-independent form allows us to find the eigenvectors and eigenvalues of the Hamiltonian.

This procedure of quantization is the means by which the classical equations of a system converted to quantum equations. It has been used over and over, in fields ranging from the motion of particles to quantum electrodynamics; there have even been (not so successful) attempts to quantize Einstein's theory of gravity. As we saw in one simple case, the procedure guarantees that the motion of expectation values is closely related to classical motion.

All of this raises a "chicken and egg" question: Which comes first—classical theory or quantum theory? Should the logical starting point of physics be classical or quantum mechanical? I think the answer is obvious. Quantum mechanics is the real description of nature. Classical mechanics, while beautiful and elegant, is nevertheless an approximation. Roughly speaking, it holds true when wave functions maintain their shape as packets. Sometimes, we're lucky and the quantum theory of a system can be guessed—and that's all it is, a guess—by starting with a familiar classical system and quantizing it. Sometimes this works. The quantum motion of electrons, deduced from the classical mechanics of particles, is a case in point. Quantum electrodynamics, deduced from Maxwell's equations, is another. But sometimes there is no classical theory to use as a starting point. The spin of a particle has no real classical counterpart. And the quantization of general relativity has largely failed. Quantum theory is probably much more fundamental than classical theory, which generally should be understood as an approximation.

That being said, I will now continue to quantize the motion of particles, but this time incorporating the effects of forces.